



computational_err...

Scientific Computation

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Lecture 6 - Floating Point Summary

Lecture Outline

- Computational error
- Absolute vs. relative error
- Computational error example
- The floating point mindset
- The story so far

Approximating the First Derivative

$$f \in C^2[a, b]$$

Let $f(x)$ be a function with continuous second derivative.

$$f(x+h) = f(x) + f'(x) \cdot h + \frac{f''(\xi)}{2!} h^2, \quad \xi \in [x, x+h]$$

$$\rightarrow f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{f''(\xi)}{2!} h$$

Therefore, if we use $f'(x) \approx \frac{f(x+h) - f(x)}{h}$,

then our truncation error is

$$-\frac{f''(\xi)}{2} h$$

Computational Error

Computational error = (rounding error) + (truncation error)

Rounding error: difference between the result produced using finite-precision arithmetic (e.g., IEEE Float32) and exact arithmetic

Truncation error: difference between the true result and the result that would be produced by an (inexact) algorithm using exact arithmetic

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2nd Order Truncation Error

Example 5: Approximate $f'(x)$ with $\frac{1}{h}(f(x+h) - f(x))$.

$$E_2 = -\frac{1}{2} f''(\xi) \quad \text{for } \xi \in [x, x+h]$$

⇒ no mention of FP

Computational Error Example

Example 6: Consider evaluating $\frac{1}{h}(\underbrace{f(x+h)}_{\approx f'(x)} - \underbrace{f(x)})$ in finite-precision arithmetic.

In finite precision (e.g. FP), we do not compute $f(x+h)$ exactly, we denote the computed value by f_1 . Then $\underline{f_1} = f(x+h) + \underline{\delta_1}$ for some δ_1 .

Similarly, we compute $\underline{f_2}$ and some $\underline{\delta_2}$:

$$\underline{f_2} = f(x) + \underline{\delta_2}$$

$f(x)$ & $f(x+h)$ are $\mathbb{R}$ -valued (ideal),

f_1, f_2 are what we actually compute.

Computational Error Example

Example 6 cont.: The we approximate $f'(x)$ by:

$$\frac{f_1 - f_2}{h} = \frac{f(x+h) - f(x)}{h} + \frac{\delta_1 - \delta_2}{h}$$

Ignoring the error in the subtraction and division in $(f_1 - f_2)/h$, the total computational error is:

$$\underbrace{f'(x)}_{\text{true value}} - \underbrace{\frac{f_1 - f_2}{h}}_{\text{approx.}} = \frac{\cancel{f(x+h)} - \cancel{f(x)}}{h} - \frac{1}{2} f''(\xi) h - \frac{\cancel{f(x+h)} - \cancel{f(x)}}{h} - \frac{\delta_1 - \delta_2}{h}$$

$$= \underbrace{-\frac{1}{2} f''(\xi) h}_{\text{(truncation)}} - \underbrace{\frac{\delta_1 - \delta_2}{h}}_{\text{(roundoff)}}$$

Denote by M the max. of $|f''(x)|$ for $x \in [x, x+h]$, and $|\delta_1|, |\delta_2| \leq \epsilon_{\text{mach}}$

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$$|a + b| \leq |a| + |b|$$

Computational Error Example

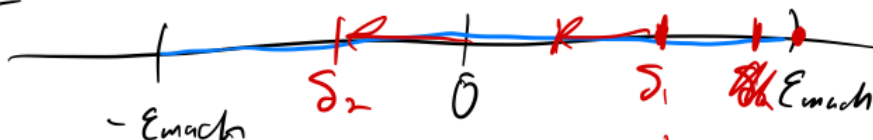
Example 6 cont.: Then

$$\left| f'(x) - \frac{f_1 - f_2}{h} \right| = \left| -\frac{1}{2} f''(\xi) h - \frac{\delta_1 - \delta_2}{h} \right|$$

$$\stackrel{\text{(triangle ineq.)}}{\leq} \left| \frac{1}{2} f''(\xi) h \right| + \left| \frac{\delta_1 - \delta_2}{h} \right|$$

$$\stackrel{\text{(triangle ineq.)}}{\leq} \frac{1}{2} M h + \frac{2 \epsilon_{\text{mach}}}{h}$$

$$\left. \begin{array}{l} |\delta_1| \leq \epsilon_{\text{mach}} \\ |\delta_2| \leq \epsilon_{\text{mach}} \end{array} \right\} \Leftrightarrow \begin{array}{l} \delta_1 \geq -\epsilon_{\text{mach}} \\ \delta_1 \leq \epsilon_{\text{mach}} \end{array}$$



$$|\delta_1 + \delta_2| \leq 2 \epsilon_{\text{mach}}$$

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Absolute vs. Relative Error

For some exact result y and some approximation $\tilde{y}$, the **absolute error** is $|y - \tilde{y}|$

The **relative error** is $\frac{|y - \tilde{y}|}{|y|}$

Recall our result about FP rounding: $fl(x) = x(1 + \delta)$, where $\delta \leq u$.

Approximation $|fl(x) - x| = |x\delta| \leq |x|u$. Since relative error

rearranging, $\left| \frac{o}{x} \right| = |o| \leq u$, so o is a **relative** error

We saw with cancellations that relative error can be terrible even when absolute error is small:

$$\left| \frac{fl(x) - fl(y) - (x - y)}{x - y} \right| = \left| \frac{x\delta_1 - y\delta_2}{x - y} \right|$$

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Computational Error

Computational error = (truncation error) + (rounding error)

$$g(h) = \frac{1}{2}Mh + 2\frac{\epsilon_{\text{mach}}}{h}$$

Comes from truncated series
(but assumes exact arithmetic)

Comes from floating point
representations

Let's take a closer look!

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The Floating Point Mindset

In their classic text *Matrix Computations*, Golub and van Loan present the following maxims for aspiring “floating point thinkers”:

1. Order is important
2. Larger may mean smaller
3. A math book is not enough

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Lecture Outline

- IEEE 754 visualizer
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Scientific Endeavour



Mathematical Model



Taylor series

Truncation error

Roundoff error

Cancellations



Results



IEEE 754

Computation

How do FP representations fit in?

